What if the soundboard were the only thing that mattered?

GPU-Accelerated Soundboard Design with Physical Fabrication and Acoustic Verification

Co-Development Proposal: Industry + University

PIANOID — Real-Time GPU-Accelerated Physical Modeling Piano Synthesizer

The Problem

Why do some pianos sing while others stay dull — and why can't we answer this systematically?

Systematic soundboard design is economically and methodologically impossible with current tools. A concert grand has 230 strings coupled through a soundboard with thousands of resonance modes. Evaluating a single design change requires weeks of skilled labor and thousands in materials.

- "Killer octave" — poor sustain in 5th–6th octave due to modal density and impedance mismatch. The mechanism is understood; the **design solution is not**
- Design knowledge exists as **intuition accumulated over decades**, not as transferable, quantifiable parameters

Research question:
Can a closed-loop methodology — FEA design, real-time acoustic preview, physical fabrication, and quantified modal verification — make soundboard design systematic and reproducible?

The core experiment: **A/B synthesis** — FEA-predicted modes vs. physically measured modes on the same engine, isolating model accuracy from synthesizer fidelity. If the predicted and measured soundboards converge perceptually, the design loop is validated.

Research Landscape

Group	Focus	Key Contribution
Chabassier, Chaigne (Inria/ENSTA)	Full piano FE simulation	Energy-preserving schemes for string + soundboard + air (JASA 2013)
MAESSTRO (ANR consortium)	Soundboard CAD tool	FEA modal synthesis; string-soundboard coupling; offline single-note (Acta Acustica 2022)
Ege, Boutillon (Polytechnique)	Experimental modal analysis	High-res soundboard characterization at manufacturing stages (2018)
Välimäki, Bank (Aalto/Budapest)	Real-time piano models	Efficient digital waveguide / IIR with psychoacoustic tuning
Modartt Pianoteq	Commercial physical model	~1000 modes/note, proprietary, no design pipeline
Weinreich (U. Michigan)	String coupling theory	Coupled polarization and double decay (JASA 1977)

All prior work operates within the conventional piano's constraints. No existing project treats the soundboard as an independent design variable, and no work closes the loop from FEA-derived modes through real-time synthesis to quantified physical verification.

What Already Exists: The Pianoid Platform

Working System — demonstrated at NAMM Show and in live concert performances

- **Synthesis algorithm** — full string set modelled with finite-difference wave equation solver, 256 soundboard modes coupled to all strings simultaneously, real-time on NVIDIA 4070 Ti with 30–50% safety margin
- **String-mode coupling network** — each soundboard mode is energised by string motion and returns energy back, creating continuous bidirectional exchange between all strings and all modes. Sympathetic resonance, pedal effects emerge naturally
- **Patented excitation system** — 8-channel electromechanical actuators, tested and proven on a real piano soundboard
- **Modal measurement pipeline** — multi-band ESPRIT extraction with automated mode tracking across 78+ bridge positions

What This Proposal Builds

- **FEA import pipeline** — extract string-mode coupling matrices from FEA eigenmode output
- **Mode capacity: 256 → 2048** — already in pipeline. Flatten coupling coefficients for higher modes
- **String discretisation improvement** — higher-resolution finite-difference string model
- **Enhanced hammer excitation model** — more accurate initial transient
- **Soundboard fabrication pipeline** — rapid prototyping, material testing, encasing design
- **Perceptual validation protocol** — ABX (forced-choice perceptual comparison) listening tests, comparative evaluation

Piano Modelling State of the Art — And Its Gaps

Capability	MAESSTRO (ANR)	Pianoteq (Commercial)	Pianoid (current)	Proposed System
Synthesis speed	16 min / note	Real-time	Real-time, 48 kHz	Real-time, 48 kHz
Polyphony	Single note	Full	Full set of strings	Full set of strings
Multi-string coupling	No (single note)	Proprietary	256 modes	2048 modes
Soundboard from FEA	Yes (core feature)	Hand-tuned	Measured only	FEA-derived
Modal measurement	Not integrated	Not integrated	ESPRIT pipeline	ESPRIT pipeline
Design verification	Not addressed	Not addressed	Not yet	A/B synthesis
Unconstrained soundboard	Conventional geometry	Conventional geometry	8-ch prototype	Multi-point excitation

Blue = proposed capabilities that this project delivers.

How We Enable the Loop

Closing the design loop requires two capabilities that no existing tool provides:

Real-Time Polyphonic Preview

GPU modal synthesis with full string-mode coupling lets a pianist **play** the designed soundboard — not listen to single offline notes. Design changes are heard instantly. Iteration takes minutes, not weeks.

Patented Excitation System

Electromechanical actuators replace physical strings entirely. No permanent tension, no iron plate, no frame. Soundboards are **swappable** — free mounting with no stress. A new design can be tested acoustically in hours.

Together

These make the FEA → preview → fabricate → measure → verify cycle practically executable.

Beyond the loop

Freeing the soundboard from its structural role also opens the door to designs that no conventional piano could accommodate.

The Closed Loop

HASE 1 Design → PHASE 2 GPU Preview → PHASE 3 Fabrication → PHASE 4 Measurement → PHASE A/B Evalu

← Below threshold? Revise FEA model, iterate from Phase 1

Loop closure: quantified modal correlation (frequency, MAC, damping) defines go/no-go. A/B perceptual test validates convergence.

Phases 1–2: Design & Preview

FEA eigenvalue analysis (COMSOL/Abaqus). Per mode: **frequency, quality factor, modal mass, mode shape amplitudes at excitation positions**. Mode shapes define the string-mode coupling matrices. Import into Pianoid for real-time polyphonic preview.

Phase 3: Fabrication

Reduced-scale test boards for rapid iteration. Material options: spruce, plywood, composites. Carbon fiber ribs are **a hypothesis to test**, not a proven approach.

Phase 4: Modal Verification

Multi-channel ESPRIT + automated mode tracking. FEA-vs-measured: frequency, MAC (Modal Assurance Criterion), damping.

Phase 5: A/B Synthesis & Evaluation

The core experiment: same engine, FEA modes vs. measured modes. Isolates model accuracy from synthesizer fidelity. ABX listening tests, expert musician assessment.

Co-Development: Industry + University (partner institution TBD)

Pianoid (Industry Partner) Brings

- **GPU synthesis engine** — real-time modal synthesis with string-mode coupling (working, demonstrated)
- **Patented excitation system** — 8-channel actuator prototype, hardware design, signal routing
- **Modal measurement pipeline** — ESPRIT extraction, automated mode tracking, FEA comparison tools
- **FEA import tool development** — software bridge from FEA eigenmode output to engine parameters
- **Engine and algorithm R&D** — mode capacity, string discretisation, hammer model improvements

University Partner Brings

- **FEA expertise** — accurate soundboard modeling: orthotropic material characterization, rib/bridge geometry, boundary conditions, mesh convergence
- **Music perception science** — ABX test design, psychoacoustic evaluation methodology, statistical analysis
- **Prototyping and testing facilities** — vibration lab, anechoic chamber, accelerometer arrays, CNC
- **Acoustic radiation modelling** — sound field prediction at the listening point; post-processing from modal data with potential upgrade to coupled structural-acoustic FEA
- **Credentialing in acoustics community** — access to peer-reviewed venues, conference networks

IP: commercial technology retained by Pianoid. Academic publications and modal datasets fully open. Specific terms negotiated jointly.

Why Now?

Technology Convergence

- **GPU compute is crossing the threshold** — real-time synthesis with finite-difference string modelling and full string-mode coupling is feasible for the first time
- **MAESSTRO laid the FEA groundwork** — soundboard modal synthesis is proven; the missing piece is real-time performance and a verification loop
- **Modal analysis tools are mature** — ESPRIT and related methods now extract hundreds of modes reliably

Platform Readiness

- **Pianoid is demonstrated** — patented excitation system, engine, and measurement pipeline shown at NAMM and in live concert performances
- **The unconstrained design concept is now executable** — the required combination of real-time synthesis, FEA import, modal measurement, and excitation hardware has only recently become available

Growing Demand

- **Sustainable materials** — new candidate soundboard materials need acoustic evaluation tools
- **Digital twin methodology** — established in aerospace and automotive; extending it to acoustic instruments is timely

Timeline and Risks

Timeline (12 months)

Phase	Months	Deliverable
Mode capacity scaling	1–2	2048-mode validation
String & hammer model improvements	1–3	Upgraded string + excitation model
FEA import tool	1–3	Working pipeline
FEA soundboard design	2–5	Modal basis, 2+ designs
GPU preview & iteration	4–6	Playable virtual instruments
Prototype fabrication	4–8	Reduced-scale test boards
Measurement & verification	7–9	FEA-vs-measured data
A/B evaluation & performance	8–11	Playable prototype, ABX
Publication & dissemination	10–12	Papers, dataset

Key Risks

Algorithm improvements fail

Mode scaling, string discretisation, or hammer model enhancements exceed real-time budget or prove unstable. **Fallback:** current algorithm stays as-is — already demonstrated in live performances and proven adequate for the research.

Poor FEA-to-measured correlation

First prototype modes don't match. **Mitigation:** start simple. Discrepancies inform refinement — this is the research.

Prototype acoustic validity

Non-traditional materials as proxies. **Mitigation:** spruce baseline alongside experimental materials.

Partnership timeline

Academic schedules, facility access. **Mitigation:** parallel workstreams; industry work proceeds independently months 1–3.

Expected Outcomes

Scientific

- **First systematic study** of soundboard acoustics freed from string tension, case geometry, and fixed bridge constraints
- **A/B synthesis results** — controlled comparison of FEA vs. measured modes on the same engine
- **Design guidelines** — quantified sensitivity of acoustic features to structural parameters
- **Open modal dataset** — published characterizations at successive construction stages

Publications

- Unconstrained soundboard design methodology (Acta Acustica / JASA)
- Mode tracking methodology
- Hybrid instrument: concept and perceptual evaluation

Technological

- **Playable prototype** — fabricated soundboard with excitation system, live acoustic performance
- **Software pipeline** — integrated FEA-to-Pianoid import workflow
- **Modal measurement toolkit** — automated ESPRIT + tracking for instrument characterization

Industrial

- Proof of concept for **computer-aided soundboard design** with real-time preview
- Pathway to **sustainable materials evaluation**
- Foundation for **novel acoustic instrument design** — soundboards unconstrained by piano geometry, with freely placed excitation and programmable string-mode coupling

What does a soundboard sound like when it is designed purely for sound?

Arbitrary geometry. Any material. Excitation at any point.
No tension. No frame. No constraints inherited from strings.
Every design decision driven by acoustic goals alone.

Now we have the tools to find out — and to design instruments that could never have existed before.

PIANOID